DIFFERENTIAL EQUATIONS

LECTURE NOTES
WEEK XI

SERIES SOLUTIONS

We have linear but not necessarily constant coefficient equation. Then, we try to find a Taylor series approximation to each of its solutions near a given point. We'll work 2nd order equations (everything can be generalized as well to any order)

REVIEW OF POWER SERIES

(1) Any series of the form

$$\sum_{n=0}^{\infty} a_n (x-x_0)^n \text{ is called a Power series}$$

A power series converges at a point x if

$$\lim_{N\to\infty} \sum_{n=0}^{N} a_n (x-x_0)^n \text{ exists for that } x.$$

For the convergence: Ratio test

$$\lim_{n\to\infty} \left| \frac{a_{n+1}(x-x_0)^{n+1}}{a_n (x-x_0)^n} \right| = \lim_{n\to\infty} |x-x_0| \underbrace{\left| \frac{a_{n+1}}{a_n} \right|}_{L} = L|x-x_0|$$

Ratio test says

(a) If $L|x-x_0|<1$ series converges absolutely $\Rightarrow$ converges

$|x-x_0| < \frac{1}{L} = \rho$ $\to$ radius of convergence

$x_0-\rho < x < x_0+\rho$

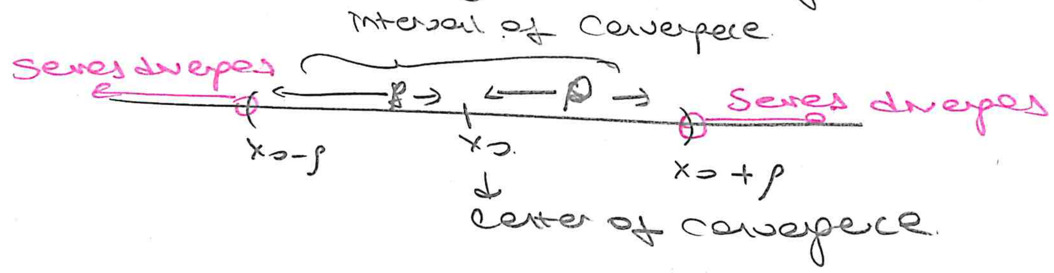
$I = (x_0-\rho, x_0+\rho) \to$ interval of convergence $\rho >$

(b) If $L|x-x_0| > 1 \Rightarrow$ series diverges.

(c) If $L|x-x_0| = 1 \Rightarrow$ test fails

$x = x_0 + \rho$ and $x = x_0 - \rho$ End points of I

Use another way for checking convergence at end point



For a power series (1) $L = \infty$ $\rho = \frac{1}{L} = 0$ power series converges at only x_0 center of convergence

(2) $L = 0$ $\rho \to \infty$ $I = (-\infty, \infty)$ series converges $\forall x \in \mathbb{R}$ $I = [x_0, x_0]$

(3) $0 < L < \infty$ $\rho = \frac{1}{L}$ is a finite number

Power series converges in $I = (x_0 - \rho, x_0 + \rho)$

But check end points $x = x_0 \mp \rho$ seperately

(2) For $|x - x_0| < \rho$, the power series defines a function of f $f(x) = \sum_{n=0}^{\infty} a_n (x-x_0)^n$ $|x - x_0| < \rho$

This function is infinitely differentiable, and in fact the series can be differentiated term by term for $|x-x_0| < \rho$. That is,

$$f'(x) = \sum_{n=0}^{\infty} a_n \cdot n \, (x-x_0)^{n-1} = \sum_{n=1}^{\infty} a_n \cdot n \, (x-x_0)^{n-1}$$

$$f''(x) = \sum_{n=1}^{\infty} a_n \, n(n-1)(x-x_0)^{n-2} = \sum_{n=2}^{\infty} a_n \cdot n(n-1) x^{n-2}$$

$$f(x) = a_0 + a_1(x-x_0) + a_2(x-x_0)^2 + \ldots$$

(3) If $a_n = \frac{f^{(n)}(x_0)}{n!}$ $n = 0, 1, 2, \ldots$ then the power series

$$\sum_{n=0}^{\infty} a_n (x-x_0)^n = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x-x_0)^n$$ is called **Taylor series** expansion of $f(x)$ around $x = x_0$

(4) A function f that has a Taylor series expansion about $x = x_0$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n$$

with a radius of convergence $|x-x_0| < \rho$ ($\rho > 0$) is said to be **analytic** at $x = x_0$. If f, g are analytic, then $f \mp g$, $f \cdot g$, f/g ($g \neq 0$) are analytic at $x = x_0$.

(5) **Index shift** $\underbrace{\sum_{n=m} a_n x^n}_{a_m x^m + a_{m+1} x^{m+1} + \cdots} = \underbrace{\sum_{n=0} a_{n+m} x^{n+m}}_{a_m x^m + a_{m+1} x^{m+1} + \cdots}$ m integer.

$$\sum_{n=2}^{\infty} 2^n n! = \sum_{n=0}^{\infty} 2^{n+2}(n+2)! = \sum_{n=1} 2^{n+1}(n+1)!$$

I SERIES SOLUTION NEAR AN ORDINARY POINT

Definition $P(x)y'' + Q(x)y' + R(x)y = 0$ (1)

Assume $P(x), Q(x), R(x)$ are analytic functions in some interval

$|x - x_0| < \rho$.

A point x_0 such that $P(x_0) \neq 0$ is called an ordinary point. On the other hand, if $P(x_0) = 0$ then x_0 is called a singular point. of equation (1).

If we put Eq (1) in the form

$$y'' + \underbrace{\frac{Q(x)}{P(x)}}_{p(x)} y' + \underbrace{\frac{R(x)}{P(x)}}_{q(x)} y = 0 \qquad (2)$$

Then, Eq (2) has a solution if $p(x)$ and $q(x)$ are continuous in an interval around x_0. But, this just means that $P(x_0) \neq 0$ since we had required P, Q, R to be analytic in an interval around x_0.

A point x_0 is called an ordinary point of Eq (2) if

$$p(x) = \frac{Q(x)}{P(x)} \quad \text{and} \quad q(x) = \frac{R(x)}{P(x)} \quad \text{are analytic function,}$$

that has Taylor series expansions around x_0, in an interval $|x - x_0| < \rho$. Otherwise, it is a singular point

Thm Let $P(x), Q(x), R(x)$ be analytic functions for $|x-x_0|<\rho$

Suppose x_0 is an ordinary point of

$$P(x)y'' + Q(x)y' + R(x)y = 0$$

then the general solution is

$$y = \sum_{n=0}^{\infty} a_n (x-x_0)^n = a_0 y_1(x) + a_1 y_2(x)$$

$$= a_0 \left[1 + \sum_{n=1}^{\infty} a_n (x-x_0)^n \right]$$

$$+ a_1 \left[(x-x_0) + \sum_{n=2}^{\infty} b_n (x-x_0)^n \right]$$

where a_0, a_1 are arbitrary, and y_1 and y_2 are linearly independent series solutions, that are analytic at x_0.

Further, the radius of convergence of each of the series solutions of y_1 and y_2 is at least as large as the minimum of the radius of convergence of the series for $\frac{Q(x)}{P(x)}$ and $\frac{R(x)}{P(x)}$

Ex Find a lower bound for ρ about x_0! $x_0 = 4$, $x_0 = -4$, $x_0 = 0$

(a) $\underbrace{(x^2-2x-3)}_{P(x)} y'' + xy' + 4y = 0$

$P(x) = x^2 - 2x - 3 = 0$ if $x = 3$ $x = -1$ $\Rightarrow$ $x = -1, x = 3$ are singular points and all other points are ordinary

$$\frac{Q}{P} = \frac{x}{x^2-2x-3} \qquad \frac{R}{P} = \frac{4}{x^2-2x-3}$$

$x_0 = 4$ $\sum_{n=0}^{\infty} a_n (x-4)^n$ converges at least for $|x-4| < \underbrace{|4-3|}_{\text{distance between 4 and nearest singular}} = 1$

$x_0 = -4$ $\sum_{n=0}^{\infty} a_n (x+4)^n$ converges at least for $|x+4| < \underbrace{|-4+1|}_{\text{distance between -4 and nearest singular}} < 3$

$x_0 = 0$ $\sum_{n=0}^{\infty} a_n x^n$ converges at least for $|x-0| < \underbrace{|0+1|}_{\text{0 and nearest singular -1.}} = 1$

(b) $(1+x^2)y''-y=0$

$P(x)=1+x^2=0 \iff x=\mp i$ singular points

$\underline{x_0=0}$ $\sum_{n=0}^{\infty} a_n x^n$ converges $\underline{\text{at least}}$ for $|x-0|=|x|=|0-i|=\sqrt{0+(-1)^2}=1$

$\rho\geq 1$

$\underline{x_0=1}$ $\sum_{n=0}^{\infty} a_n (x-1)^n$ Converges $\underline{\text{at least for}}$ $(x-1)=|1-i|=\sqrt{1+(-1)^2}=\sqrt{2}$

Why we say at least for example. this question has solution $\rho\geq 2$.
$y(x)=C_1\cos(\operatorname{arctan}x)+C_2\sin(\operatorname{arctan}x)$ analytic for all $x\in\mathbb{R}$

(c) $(x-5)(x^2+9)y''-y=0$

$P(x)=(x-5)(x^2+9)=0 \Rightarrow x=5$ $x=\mp 3i$ singular points

Other points are ordinary points

$\underline{x_0=0}$ $\sum_{n=0}^{\infty} a_n x^n$ converges at least for $|x|<|0-5|=5$. Choose its min

$|x|<|0-3i|=3$.

So $\rho\geq 3$

Ex Find the general solution of Airy Equation.

$$y''-xy=0$$

$P(x)=1$, $Q(x)=0$, $R(x)=-x$ are all analytic and $P(0)=1$

So $x_0=0$ is an ordinary point. So let $y=\sum_{n=0}^{\infty} a_n x^n$, then

$y'=\sum_{n=1}^{\infty} n a_n x^{n-1}$, $y''=\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2}$. Substitute in D.E

$y''-xy=0.$

$$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - x\cdot\sum_{n=0}^{\infty} a_n x^n=0.$$

$$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - \sum_{n=0}^{\infty} a_n x^{n+1}=0$$

" First equate power of x, then start from the same index"

$$\sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n - \sum_{n=1}^{\infty} a_{n-1}x^n=0.$$

$$2\cdot 1\, a_2 + \sum_{n=1}^{\infty}\left[(n+2)(n+1)a_{n+2}-a_{n-1}\right]x^n=0.$$

$\Rightarrow 2a_2 = 0 \quad a_2 = 0$

$\Rightarrow (n+2)(n+1)\, a_{n+2} - a_{n-1} = 0 \qquad n \geq 1$

$$a_{n+2} = \frac{a_{n-1}}{(n+2)(n+1)} \qquad n \geq 1$$ "recursive relation"

$\underline{n=1}$ $\quad a_3 = \frac{a_0}{3.2}$ $\qquad \underline{n=2}$ $\quad a_4 = \frac{a_1}{4.3}$ $\qquad \underline{n=3}$ $\quad a_5 = \frac{\overset{0}{\overset{\|}{a_2}}}{5.3} \Rightarrow a_5 = a_8 = a_{11} = \ldots = a_{3n+2} = 0 \quad n \geq$

$\underline{n=4}$ $\quad a_6 = \frac{a_3}{6.5} = \frac{a_0}{6.5.3.2}$ $\qquad \underline{n=5}$ $\quad a_7 = \frac{a_4}{7.6} = \frac{a_1}{7.6.4.3}.$

This way, we obtain.

$$a_{3n} = \frac{a_0}{3n(3n-1)(3n-3)\cdots 3.2} \qquad a_{3n+1} = \frac{a_1}{(3n+1)(3n)(3n-2)\cdots 4.3}$$

$$a_{3n+2} = 0$$

where a_0, a_1 are arbitrary. Then, the general solution.

$$y(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \cdots$$

$$= \sum_{n=0}^{\infty} a_{3n} x^{3n} + \sum_{n=0}^{\infty} a_{3n+1} x^{3n+1} + \sum_{n=0}^{\infty} \underset{0}{a_{3n+2}}\, x$$

$$= a_0 + \sum_{n=1}^{\infty} \frac{a_0}{3n(3n-1)(3n-3)\cdots 3.2} x^{3n}$$

$$+ a_1 x + \sum_{n=1}^{\infty} \frac{a_1}{(3n+1)3n\underset{(3n-1)}{(3n-2)}(3n-3)\cdots 4.3} x^{3n+1}$$

$$y(x) = a_0 \underbrace{\left[1 + \sum_{n=1}^{\infty} \frac{x^{3n}}{3n(3n-1)(3n-3)\cdots 3.2} \right]}_{y_1(x)}$$

$$+ a_1 \underbrace{\left[x + \sum_{n=1}^{\infty} \frac{x^{3n+1}}{(3n+1)(3n)\underset{3n-1}{(3n-2)}\cdots 4.3} \right]}_{y_2(x)}$$

Ex 7 Solve the I.V.P

$$y'' - xy = 0 \quad , y(0) = 5 \, , \; y'(0) = -4$$

By previous example

$$y(0) = 5 \quad \Rightarrow \quad y(0) = \boxed{a_0 = 5}$$

$$y'(x) = a_0 \sum_{n=1}^{\infty} \frac{3n \, x^{3n-1}}{3n(3n-1) \cdots 3.2} + a_1 + a_1 \sum_{n=1}^{\infty} \frac{(3n+1) x^{3n}}{(3n+1) 3n \cdots 4.3}$$

$$y'(0) = \boxed{a_1 = -4}$$

Thus, $y(x) = 5 y_1(x) - 4 y_2(x)$

Ex 7 Solve the I.V.P $y'' - xy = 0 \quad y(-1) = 0 , \; y'(-1) = 1$

Note In this case the initial point is $x_0 = -1$. Hence, we want an expansion in powers of $x - x_0 = x+1$

$x_0 = -1$ is an ordinary point, so let $y = \sum_{n=0}^{\infty} a_n (x+1)^n$, then

$$y' = \sum_{n=1}^{\infty} n a_n (x+1)^{n-1} , \quad y'' = \sum_{n=2}^{\infty} n(n-1) a_n (x+1)^{n-2}$$

Subs in D.E.

$$\sum_{n=2}^{\infty} n(n-1) a_n (x+1)^{n-2} - x \sum_{n=0}^{\infty} a_n (x+1)^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1) a_n (x+1)^{n-2} - \underbrace{[(x+1) - 1]}_{x} \sum_{n=0}^{\infty} a_n (x+1)^n = 0$$

$$\sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} (x+1)^n - \underbrace{\sum_{n=0}^{\infty} a_n (x+1)^{n+1}}_{\sum_{n=1}^{\infty} a_{n-1} (x+1)^n} + \sum_{n=0}^{\infty} a_n (x+1)^n = 0$$

$$2a_2 + a_0 + \sum_{n=1}^{\infty} [(n+2)(n+1) a_{n+2} - a_{n-1} + a_n](x+1)^n = 0$$

$$\boxed{a_{n+2} = \frac{a_{n-1} - a_n}{(n+2)(n+1)}} \quad n \geq 1 \quad \text{"recursive formula"}$$

$2a_2 + a_0 = 0$

$a_2 = -\frac{a_0}{2}$

$n=1 \quad a_3 = \dfrac{a_0 - a_1}{3.2} \quad = a_0/2$

$n=2 \quad a_4 = \dfrac{a_1 - a_2}{4.3} = \dfrac{a_1}{12} + \dfrac{a_0}{24}$

$n=3 \quad a_5 = \dfrac{a_1}{80} - \dfrac{a_0}{160}$

$$y = \sum_{n=0}^{\infty} a_n (x+1)^n = a_0 + a_1(x+1) + a_2(x+1)^2 + a_3(x+1)^3 + a_4(x+1)^4 + \cdots$$

$$= a_0 + a_1(x+1) + \left(-\frac{a_0}{2}\right)(x+1)^2 + \left(\frac{a_0}{6} - \frac{a_1}{6}\right)(x+1)^3 + \left(\frac{a_1}{12} + \frac{a_0}{24}\right)(x+1)^4$$

$$y = a_0 \left[1 + \frac{1}{2}(x+1)^2 + \frac{1}{6}(x+1)^3 + \frac{1}{24}(x+1)^4 - \frac{1}{160}(x+1)^5 + \cdots\right]$$

$$+ a_1 \left[(x+1) - \frac{1}{6}(x+1)^3 + \frac{1}{12}(x+1)^4 + \frac{1}{80}(x+1)^5 + \cdots\right]$$

$y(-1) = 0 \quad \Rightarrow a_0 = 0$

$y'(x) = a_1 - a_0(x+1) + 3\left(\frac{a_0}{6} - \frac{a_1}{6}\right)(x+1)^2 + \cdots$

$y'(-1) = a_1 = 1 \quad$ So $\quad \boxed{y(x) = y_2(x)}$

Or Use substitution. $t = x+1 \Rightarrow x = t-1$

$y'' - xy = 0$. Transform it for $t = x+1$

$x = t-1$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{dy}{dt}$$

(with $\frac{dt}{dx} = 1$)

$$\left(\frac{d^2y}{dx^2}\right) = \frac{d}{dt}\left(\frac{dy}{dt}\right)\frac{dt}{dx} = \frac{d^2y}{dt^2}$$

$y_{tt} - (t-1) \cdot y = 0$

Write series representation around $t=0$, i.e. $y(t) = \sum_{n=0} a_n t^n$

$$\sum_{n=2}^{\infty} n(n-1) a_n t^{n-2} - (t-1) \sum_{n=0} a_n t^n = 0$$

$$\sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} t^n - \sum_{n=1}^{\infty} a_{n-1} t^n + \sum_{n=0} a_n t^n = 0$$

$$2a_2+a_0+\sum_{n=1}^{\infty}\left[(n+2)(n+1)a_{n+2}-a_{n-1}+a_n\right]t^n=0$$

$a_2=-\frac{a_0}{2}$ $\quad (n+2)(n+1)a_{n+2}=a_{n-1}-a_n \quad n\geq 1.$

$$\underbrace{\overbrace{a_{n+2}=\frac{a_{n-1}-a_n}{(n+2)(n+1)}}}$$ same formula with ⊛

Ex Legendre equation of order one

$$(1-x^2)y''-2xy'+2y=0.$$

Singular points $1-x^2=0 \Rightarrow x=\mp 1$. All other points are ordinary.

let's solve it around $x_0=0$

$$y=\sum_{n=0}^{\infty}a_nx^n \quad y'=\sum_{n=1}^{\infty}n.a_nx^{n-1},\quad y''=\sum_{n=2}^{\infty}n(n-1)a_nx^{n-2}$$

$$(1-x^2)\sum_{n=2}^{\infty}n(n-1)a_nx^{n-2}-2x\sum_{n=1}^{\infty}na_nx^{n-1}+2\sum_{n=0}^{\infty}a_nx^n=0.$$

$$\underbrace{\sum_{n=2}^{\infty}n(n-1)a_nx^{n-2}}_{\sum_{n=0}^{\infty}(n+2)(n+1)a_{n+2}x^n}-\sum_{n=2}^{\infty}n(n-1)a_nx^n-2\sum_{n=1}^{\infty}na_nx^n+2\sum_{n=0}^{\infty}a_nx^n=0$$

$$2.1a_2+3.2a_3x-\cancel{2a_1x}+2a_0+\cancel{2a_1x}+\sum_{n=2}^{\infty}\{(n+2)(n+1)a_{n+2}-n(n-1)a_n$$

$$-2na_n+2a_n\}x^n=0$$

$2a_2+2a_0=0$ $\quad 6a_3=0$

$\boxed{a_2=-a_0}$ $\quad \boxed{a_3=0}$

$a_{n+2}=\frac{n-1}{n+1}a_n,\quad n\geq 2$

$n=2 \quad a_4=\frac{1}{3}a_2=-\frac{a_0}{3}$

$n=3 \quad a_5=\frac{2}{4}a_3=0$

$n=4 \quad a_6=\frac{3}{5}a_4=\frac{3}{5}(-\frac{1}{3})a_0=-\frac{a_0}{5}$

$n=5 \quad a_7=\frac{4}{6}a_5=0.$

$\vdots$

$a_{2k}=-\frac{a_0}{2k-1} \quad a_{2k+1}=0 \quad k\geq 1$

$$y(x)=a_0+a_1x+a_2x^2+\cdots \Rightarrow y(x)=\sum_{k=0}^{\infty}a_{2k}x^{2k}+\sum_{k=0}^{\infty}a_{2k+1}x^{2k+1}$$

$$y(x)=a_0+\sum_{k=1}^{\infty}-\frac{a_0}{2k-1}x^{2k}+a_1x+\sum_{k=1}^{\infty}0.x^{2k+1}$$

$$y(x)=a_0\underbrace{\left[1-\sum_{k=1}^{\infty}\frac{x^{2k}}{2k-1}\right]}+a_1\underbrace{x}_{y_2(x)}$$

(II) SERIES SOLUTION NEAR A SINGULAR POINT
REGULAR

Consider $P(x)y'' + Q(x)y' + R(x)y = 0$

If $P(x_0) = 0$, then x_0 is called a singular point.

Definition 1 If both $\lim_{x \to x_0} \frac{Q(x)}{P(x)}$ and $\lim_{x \to x_0} \frac{R(x)}{P(x)}$ exists,

then $x = x_0$ is a removable singular point of equation

$$P(x)y'' + Q(x)y' + R(x)y = 0$$

Ex $(x-1)y'' + 2(x-1)^2 y' + 3(x^2-1)y = 0$

$x_0 = 1$ is a s.p since $P(x) = x-1$

Check $\lim_{x \to 1} \frac{Q(x)}{P(x)} = \lim_{x \to 1} \frac{2(x-1)}{x-1} = 2$ exists.

$\lim_{x \to 1} \frac{R(x)}{P(x)} = \lim_{x \to 1} \frac{3(x^2-1)}{(x-1)} = 6$ exists.

So, $x = 1$ is a removable singular point.

(2) Let x_0 be a singular point of

$$P(x)y'' + Q(x)y' + R(x)y = 0.$$

Suppose, $\lim_{x \to x_0} (x-x_0) \frac{Q(x)}{P(x)} = \alpha_0$ and $\lim_{x \to x_0} (x-x_0)^2 \frac{R(x)}{P(x)} = \beta_0$

exist and both finite, then x_0 is called a regular singular point. (R.S.P)

In this case, the functions defined as

$\alpha(x) = (x-x_0) \frac{Q(x)}{P(x)}$, $\beta(x) = (x-x_0)^2 \frac{R(x)}{P(x)}$ for $x \neq x_0$

$\alpha(x_0) = \alpha_0$ $\beta(x_0) = \beta_0$

and $\alpha(x_0)=\alpha_0$, $\beta(x_0)=\beta_0$ will be analytic in an interval $|x-x_0|<\rho$ around x_0. The radius of convergence ρ will be the smallest distance of x_0 to the other (real or complex) roots of $P(x)=0$. in the case when P, Q, R are polynomials.

Exm $2x(x-1)y''+(1+3x)y'+3y=0$ Find and classify singular points.

$P(x)=2x(x-1)=0 \Rightarrow x=0, x=1$ are S.Ps. All other points are ordinary points.

$x_0=0$ R.S.P?

$$\lim_{x\to 0}(x-0)\frac{1+3x}{2x(x-1)}=-\frac{1}{2}=\alpha_0 \qquad \lim_{x\to 0} x^2\cdot\frac{3}{2x(x-1)}=0=\beta_0 \text{ exist so.}$$

$x_0=0$ is a R.S.P.

That is, $\alpha(x)=x\,\dfrac{1+3x}{2x(x-1)}$ and $\beta(x)=x^2\cdot\dfrac{3}{2x(x-1)}$ are analytic at $x_0=0$.

$x_0=1$ $$\lim_{x\to 1}(x-1)\frac{(1+3x)}{2x(x-1)}=2=\alpha_0 \text{ and } \lim_{x\to 1}(x-1)^2\frac{3}{2x(x-1)}=0=\beta_0$$

Both α_0 and β_0 are finite so $x_0=1$ R.S.P.

Ex $(1-x^2)y''-2xy'+\alpha(\alpha+1)y=0$.

$p(x)=1-x^2=0$ $x=\mp 1$. S.Ps

$x_0=1$ $$\alpha_0=\lim_{x\to 1}(x-1)\frac{-2x}{(1-x^2)}=1.$$
$$\beta_0=\lim_{x\to 1}(x-1)^2\frac{\alpha(\alpha+1)}{1-x^2}=0$$
$\Big\}$ $x_0=1$ is a R.S.P.

$x_0=-1$ $$\alpha_0=\lim_{x\to -1}(x+1)\frac{-2x}{1-x^2}=1$$
$$\beta_0=\lim_{x\to -1}(x+1)^2\frac{\alpha(\alpha+1)}{1-x^2}=0$$
$\Big\}$ $x_0=-1$ is a R.S.P.

Ex $2x(x-2)^2y'' + 3xy' + (x-2)y = 0$

$p(x) = 2x(x-2)^2 = 0 \Rightarrow x=0$ and $x=2$ are S.Ps.

$x_0=0$ $\quad \lim_{x\to 0} x.\frac{3x}{2x(x-2)^2} = 0 = \alpha_0, \quad \lim_{x\to 0} x^2 \frac{x-2}{2x(x-2)^2} = 0 = \beta_0$

$\Rightarrow x_0 = 0$ is a R.S.P.

$x_0=2$ $\quad \lim_{x\to 2} (x-2)\frac{3x}{2x(x-2)^2} = \infty$ limit D.N.E $x=2$ is an irregular S.

EULER EQUATION

$L[y] = x^2y'' + \alpha xy' + \beta y = 0 \quad \alpha,\beta \in \mathbb{R}$ (*)

Note that on any interval not containing the origin the equation has a solution of the form $(x=0 \Rightarrow$ singular point$)$

$P(x)=x^2$ $R(x)=\beta$ $Q(x)=\alpha x$

$\alpha_0 = \lim_{x\to 0} x \frac{x.\alpha}{x^2} = \alpha$ $\frac{Q(x)}{P(x)}$ R.S.P

$\beta_0 = \lim_{x\to 0} x^2. \frac{\beta}{x^2} = \beta$ $\frac{R(x)}{P(x)}$

$y = c_1y_1 + c_2y_2.$

where y_1 and y_2 are linearly independent solutions.

First assume $x>0$ and try $y = x^r, r\in\mathbb{R}$ as a solution.

$L[x^r] = x^2(x^r)'' + \alpha x(x^r)' + \beta x^r = 0$

$r(r-1)x^{r-2}x^2 + \alpha r x x^{r-1} + \beta x^r = 0$

$x^r(\underbrace{r(r-1) + \alpha r + \beta}_{f(r)=0}) = 0 \Rightarrow L[x^r] = x^r f(r)$

Consider $x = e^t$ $y(x) = u(t)$ then $y' = \frac{dy}{dx} = \frac{du}{dt}\cdot\frac{dt}{dx} = \frac{u'}{x}$

$y'' = \frac{d^2y}{dx^2} = \frac{u''-u'}{x^2}$ Then (*) takes the form $u'' + (\alpha-1)u' + \beta u = 0$ with Char sol $r^2 + (\alpha-1)r + \beta = 0$ $e^{rt} = x^r$

where $f(r) = r(r-1) + \alpha r + \beta = 0$ "indicial equation"

$r^2 + (\alpha-1)r + \beta = 0$

$r_{1,2} = -\frac{(\alpha-1) \mp \sqrt{(\alpha-1)^2 - 4\beta}}{2}$

Consider $x = e^t$ $y(x)=$ then (*) takes the form

$y' = \frac{dy}{dx} = \frac{du}{dt}\cdot\frac{dt}{dx} = \frac{u'}{x}$

$y'' = \frac{u''-u'}{x^2}$

Then (*) takes the form $u'' + (\alpha-1)u' + \beta u =$

Char eq. $r^2 + (\alpha-1)r + \beta$

So for $e^{rt}, e^{\cdots}$ we have $x^r, \cdots$

We have 3 cases

① Real distinct roots: $y = c_1x^{r_1} + c_2x^{r_2}$

② Complex roots: $r_{1,2} = \lambda \mp i\mu$

$x^r = e^{r\ln x} = e^{(\lambda+i\mu)\ln x} = e^{\lambda \ln x} + e^{i\mu \ln x}$

$= x^\lambda [\cos(\mu\ln x) + i\sin(\mu \ln x)]$

③ Equal roots $r_1 = r_2 = \frac{1-\alpha}{2} = r$ $\quad y_1 = x^r = e^{r(\ln x)} =$

$\cdots = \ln x \, x^r$

(I) REAL DISTINCT ROOTS

If $f(r) = r(r-1) + \alpha r + \beta = 0$ has real roots, r_1, r_2 with $r_1 \neq r_2$

then $y_1 = x^{r_1}$ and $y_2 = x^{r_2}$ are solutions. Since,

$$W(y_1, y_2) = \begin{vmatrix} x^{r_1} & x^{r_2} \\ r_1 x^{r_1 - 1} & r_2 x^{r_2 - 1} \end{vmatrix} = (r_2 - r_1) x^{r_1 + r_2 - 1} \neq 0 \text{ for } r_1 \neq r_2 \text{ and } x > 0$$

The general solution is $\boxed{y = c_1 x^{r_1} + c_2 x^{r_2}}$ $x > 0$.

(II) EQUAL ROOTS

$r_1 = r_2 \Rightarrow \boxed{y_1 = x^{r_1}}$ Find y_2. $[L[y] = x^2 y'' + \alpha x y' + \beta y = 0$

$r_1 = r_2 \Rightarrow f(r) = (r - r_1)^2$ with $f(r_1) = 0$, $f'(r_1) = 0$ as well

Let's differentiate $L[x^r] = x^r[r(r-1) + \alpha r + \beta] = x^r f(r)$

$$\boxed{\left(\frac{d}{dx} a^x\right) = \left(a^x \ln x\right)}$$

$$\frac{\partial}{\partial r} L[x^r] = \frac{\partial}{\partial r}[x^r f(r)]$$

$$L\left[\frac{\partial}{\partial r} x^r\right] = \frac{\partial}{\partial r}[x^r (r - r_1)^2]$$

$$L[x^r \ln x] = x^r \ln x \underbrace{(r - r_1)^2}_{0} + 2x^r \underbrace{(r - r_1)}_{0}$$

If $r_1 = r_2$, $\boxed{\frac{\partial}{\partial r} L[x^r]\Big|_{r=r_1} = L[x^r \ln x]\Big|_{r=r_1}} = L[x^{r_1} \ln x] = 0$

Such that $\boxed{y_2(x) = x^{r_1} \ln x}$ $x > 0$ is a second solution. $W(x^{r_1}, x^{r_1} \ln x) = x^{2r_1 - 1}$! Hence, x^{r_1} and $x^{r_1} \ln x$ are linearly independent solutions for $x > 0$

and general solution $\boxed{y = c_1 x^{r_1} + c_2 x^{r_1} \ln x}$

III COMPLEX ROOTS

Suppose $r_{1,2} = \lambda \mp i\mu$ with $\mu \neq 0$.

$x^r = e^{r \ln x}$ when $x>0$ and r is real; then

$$x^{\lambda \mp i\mu} = e^{(\lambda \mp i\mu)\ln x} = e^{\lambda \ln x} e^{\mp i\mu \ln x} = x^{\lambda} e^{\mp i\mu \ln x}$$

$$= x^{\lambda}\left[\cos(\mu \ln x) \mp i \sin(\mu \ln x)\right] \quad x>0$$

Complex valued solutions, but we are interested in real solutions.

Then, $\boxed{y_1(x) = x^{\lambda}\cos(\mu \ln x) \text{ and } y_2(x) = x^{\lambda}\sin(\mu \ln x)}$ are two real solutions

$$W(y_1, y_2) = \mu x^{2\lambda - 1} \neq 0 \quad x>0$$

The general solution $\boxed{y = C_1 x^{\lambda}\cos(\mu \ln x) + C_2 x^{\lambda}\sin(\mu \ln x)}$

Remark Look at solutions of Euler Eqn. in the interval $x<0$

by using change of variable $x = -\varepsilon$ $\varepsilon>0$ Let $y = u(\varepsilon)$.

Then, we have

$$\frac{dy}{dx} = \frac{dy}{d\varepsilon}\cdot\frac{d\varepsilon}{dx} = -\frac{dy}{d\varepsilon} \qquad \frac{d^2y}{dx^2} = \frac{d^2y}{d\varepsilon^2}$$

Thus, Euler equation.

$$L[y] = x^2y'' + \alpha x y' + \beta y = 0 \quad \text{takes the form}$$

$$\varepsilon^2 \frac{d^2y}{d\varepsilon^2} + \alpha\varepsilon\frac{dy}{d\varepsilon} + \beta u = 0, \quad \varepsilon>0$$

which is exactly the problem we discuss above.

$$u(\varepsilon) = \begin{cases} c_1\varepsilon^{r_1} + c_2\varepsilon^{r_2} & r_1 \neq r_2 \text{ real} \\ c_1\varepsilon^{r_1} + c_2\varepsilon^{r_1}\ln\varepsilon & r_1 = r_2 \text{ real} \\ c_1\varepsilon^{\lambda}\cos(\mu\ln\varepsilon) + c_2\varepsilon^{\lambda}\sin(\mu\ln\varepsilon) & r_{1,2} = \lambda \mp i\mu \text{ complex} \end{cases}$$

Result We can combine the results for $x>0$ and $x<0$ by realizing that $|x|=x$ when $x>0$ and $|x|=-x$ when $x<0$. Thus, we need only replace x by $|x|$ in the solutions

Theorem The general solution of Euler Equation:

$$x^2y''+\alpha xy'+\beta y=0$$

in any interval not containing the origin is determined by the roots of r_1 and r_2 of the equation.

$$F(r)=r(r-1)+\alpha r+\beta=0$$

$$y=\begin{cases} c_1|x|^{r_1}+c_2|x|^{r_2} & \text{if } r_1,r_2 \text{ real and distinct roots} \\ c_1|x|^{r_1}+c_2|x|^{r_2}\ln|x| & \text{if } r_1,r_2 \text{ real and equal} \\ c_1|x|^{\lambda}\cos(\mu\ln x)+c_2|x|^{\lambda}\sin(\mu\ln|x|) & \text{where } r_{1,2}=\lambda\mp i\mu,\ \mu\ne \end{cases}$$

Remark The solutions of Euler Equation of the form

$$(x-x_0)^2y''+\alpha(x-x_0)y'+\beta y=0$$

are similar to those given in theorem. If the solution in the form $y=(x-x_0)^r$ then the general solution is obtained with x replaced by $(x-x_0)$ i.e

$$y(x)=\begin{cases} c_1|x-x_0|^{r_1}+c_2|x-x_0|^{r_2} & r_1\neq r_2\in\mathbb{R} \\ (c_1+c_2\ln|x-x_0|)\,|x-x_0|^{r_1} & r_1=r_2\in\mathbb{R} \\ c_1|x-x_0|^{\lambda}\cos(\mu\ln|x-x_0|)+c_2|x-x_0|^{\lambda}\sin(\mu\ln|x-x_0|) & r_{1,2}=\lambda\mp i\mu \end{cases}$$

Ex (1) $x^2y'' - 5xy' + 9y = 0$

$f(r) = r(r-1) - 5r + 9 = 0 \Rightarrow (r-3)^2 = 0 \Rightarrow r_1 = r_2 = 3$

$y = c_1|x|^3 + c_2|x|^3 \ln|x|$

Ex (2) $x^2y'' + 2xy' + 4y = 0$ $\quad f(r) = r(r-1) + 2r + 4 = 0 \Rightarrow r_{1,2} = -\frac{1}{2} \mp \sqrt{15}\,i$

$y = c_1|x|^{-1/2}\cos(\sqrt{15}\ln|x|) + c_2|x|^{-1/2}\sin(\sqrt{15}\ln|x|)$

Ex (3) $(x+1)^2y'' + 3(x+1)y' + 0.75y = 0$

use $t = x+1$ $\quad \frac{dt}{dx} = 1$ $\quad y = u(t)$

$\frac{dy}{dx} = \frac{du}{dt}\cdot\frac{dt}{dx} = \frac{du}{dt}$ $\quad \frac{d^2y}{dx^2} = \frac{d^2u}{dt^2}$

$t^2u'' + 3tu' + 0.75u = 0$ (Euler Eqn)

$F(r) = r(r-1) + 3r + 0.75 = 0 \Rightarrow r_1 = -3/2,\ r_2 = -1/2.$

$u(t) = c_1|t|^{-3/2} + c_2|t|^{-1/2}$

$y(x) = c_1|x+1|^{-3/2} + c_2|x+1|^{-1/2}.$

BACK TO SERIES SOLUTIONS AT REGULAR SINGULAR POINTS

$\boxed{P(x)y'' + Q(x)y' + R(x)y = 0}$ where $x = x_0$ is a regular singular point, let $x_0 = 0$ W.L.O.G [If $x_0 \neq 0$ then $x - x_0 = t$ reduces the D.E into an equation with $t = 0$ is a S.P] ($t = x - x_0$)

(1) $x\frac{Q(x)}{P(x)} = x.p(x) = \alpha(x)$ $\quad$ (2) $x^2\frac{R(x)}{P(x)} = x^2q(x) = \beta(x)$

have finite. limits as $x \to 0$ and are analytic at $x = 0$.

Thus, they have convergent power series expansions

$xp(x) = \sum_{n=0}^{\infty} p_n x^n$ $\quad x^2q(x) = \sum_{n=0}^{\infty} q_n x^n.$

on some interval $|x| < \rho$ about origin $\rho > 0$.

$P(x)y'' + Q(x)y' + R(x)y = 0$ can be written

$y'' + \underbrace{\frac{Q(x)}{P(x)}}_{p(x)} y' + \underbrace{\frac{R(x)}{P(x)}}_{q(x)} y = 0$ multiply with x^2

$$x^2y'' + x[x\,p(x)]y' + [x^2 q(x)]y = 0$$

or $\boxed{x^2y'' + x(p_0 + p_1x + \cdots + p_nx^n + \cdots)y' + (q_0 + q_1x + \cdots + q_nx^n + \cdots)y}$ (*)

If all $p_n, q_n = 0$ except possibly

$p_0 = \lim_{x\to 0} x\,p(x)$ and $q_0 = \lim_{x\to 0} x^2 q(x)$

then, the equation reduces to Euler equation.

$$\boxed{x^2y'' + p_0xy' + q_0y = 0}$$

If p_n, q_n exist, then the coefficients of eqn (*) are "Euler coefficients" times power series then it is natural to seek solution in the form of "Euler solutions" times power series.

∴ Thus, we can assume that,

$$\boxed{\begin{aligned} y &= x^r(a_0 + a_1x + a_2x^2 + \cdots + a_nx^n + \cdots) \\ &= x^r\sum_{n=0}^{\infty} a_nx^n = \sum_{n=0}^{\infty} a_nx^{n+r} \end{aligned}}$$

where $\boxed{a_0 \neq 0}$. In other words, r is the exponent of the first term in the series and a_0 is its coefficient

Ex 1 $\quad 2xy''+y'-xy=0 \quad \Rightarrow y''+\frac{1}{2x}y'-\frac{1}{x}y=0$

$x_0=0$ is a R.S.P since

$\alpha_0=\lim_{x\to 0} x\cdot\frac{1}{2x}=\frac{1}{2}$ and $\beta_0=\lim_{x\to 0} x^2\,\frac{-x}{2x}=0$, finite numbers

Let $y=\sum_{n=0}^{\infty} a_n x^{n+r}$ $\quad a_0\neq 0$ is the form of the solution

$y'=\sum_{n=0}^{\infty}(n+r)a_n x^{n+r-1}$, $y''=\sum_{n=0}^{\infty}(n+r)(n+r-1)a_n x^{n+r-2}$

$2xy''+y'-xy=0$

$2x\sum_{n=0}^{\infty}(n+r)(n+r-1)a_n x^{n+r-2}+\sum_{n=0}^{\infty}(n+r)a_n x^{n+r-1}-x\sum_{n=0}^{\infty}a_n x^{n+r}=0$

$\sum_{n=0}^{\infty}2(n+r)(n+r-1)a_n x^{n+r-1}+\sum_{n=0}^{\infty}(n+r)a_n x^{n+r-1}-\sum_{n=0}^{\infty}a_n x^{n+r+1}=0$ & equate powers of x to $n+r$.

$\sum_{n=-1}^{\infty}2(n+r+1)(n+r)a_{n+1}x^{n+r}+\sum_{n=-1}^{\infty}(n+r+1)a_{n+1}x^{n+r}-\sum_{n=1}^{\infty}a_{n-1}x^{n+r}=0$

$2r(r-1)a_0x^{r-1}+2(r+1)ra_1x^r+ra_0x^{r-1}+(r+1)a_1x^r$

$+\sum_{n=1}^{\infty}\{[2(n+r+1)(n+r)+(n+r+1)]a_{n+1}-a_{n-1}\}x^{n+r}=0$

① $[2r(r-1)+r]a_0=0$, $a_0\neq 0$ $\Rightarrow 2r(r-1)+r=0$ "indicial equation"
$2r^2-r=0$ $\quad r_1=1/2$, $r_2=0$, $r_1-r_2=1/2\notin\mathbb{Z}$ (not integer)
$r_1>r_2$.

② $[2r(r+1)+r+1]a_1=0$

③ $(n+r+1)(2(n+r)+1)a_{n+1}=a_{n-1}$ $\quad n\geq 1$

To obtain first solution $y_1(x)$

Let $r_1=1/2$, the equation ② and ③ become.

② $[2r(r+1)+(r+1)]a_1=0 \Rightarrow 3a_1=0 \Rightarrow \boxed{a_1=0}$

③ $(n+\frac{1}{2}+1)\left[2(n+\frac{1}{2})+1\right]a_{n+1}=a_{n-1}\Rightarrow a_{n+1}=\frac{a_{n-1}}{(2n+3)(n+1)}$
$n=1,2,\ldots$

This recurrence relation gives us

$$n=1, \quad a_2 = \frac{a_0}{5.2}$$

$$n=2, \quad a_3 = \frac{a_1=0}{7.3} = 0$$

$$n=3, \quad a_4 = \frac{a_2}{9.4} = \frac{a_0}{9.5.4.2}$$

$$n=4, \quad a_5 = 0$$

$$n=5, \quad a_6 = \frac{a_4}{13.6} = \frac{a_0}{13.9.5.6.4.2.}$$

$$a_{2k} = \frac{a_0}{(2.4.\cdots 2k)[5.9.13\cdots(4k+1)]} \quad k=1,2,\cdots$$

$$a_{2k+1} = 0 \quad k=0,1,2,\cdots$$

Then, $y_1 = x^r \sum_{n=0}^{\infty} a_n x^n = x^{r_1} \sum_{n=0}^{\infty} a_n x^n$

$$= x^{1/2} \left\{ a_0 + \sum_{k=1}^{\infty} a_{2k} x^{2k} + \sum_{k=0}^{\infty} \underbrace{a_{2k+1}}_{0} x^{2k+1} \right\}$$

$$y_1 = a_0 x^{1/2} \left\{ 1 + \sum_{k=1}^{\infty} \frac{x^{2k}}{(2.4\cdots(2k))\,(5.9.13\cdots(4k+1))} \right\}$$

To obtain the second solution, take $r_2=0$.

Then, ② $\Rightarrow \boxed{a_1=0}$

③ $\Rightarrow (n+1)(2n+1)a_{n+1} = a_{n-1}$

$$a_{n+1} = \frac{a_{n-1}}{(n+1)(2n+1)} \quad n \geq 1$$

$$n=1, \quad a_2 = \frac{a_0}{2.3}$$

$$n=2, \quad a_3 = \frac{a_1}{3.5} = 0$$

$$n=3, \quad a_4 = \frac{a_2}{4.7} = \frac{a_0}{2.4.3.7}$$

$$n=4, \quad a_5 = \frac{a_3=0}{5.9} = 0$$

$$n=5, \quad a_6 = \frac{a_4}{6.11} = \frac{a_0}{2.4.6\ 3.7.11}$$

$$a_{2k+1} = 0 \quad k=0,1,2,\cdots$$

$$a_{2k} = \frac{a_0}{(2.4.6\cdots 2k)(3.7.11\cdots(2k-1))} \quad k \geq 1.$$

$$y_2(x) = \sum_{n=0}^{\infty} a_n x^{n+r_2} = \sum_{n=0}^{\infty} a_n x^n = a_0 + \sum_{k=0}^{\infty} a_{2k+1} x^{2k+1} + \sum_{k=1}^{\infty} a_{2k} x^{2k}$$

$$= a_0 + \sum_{k=1}^{\infty} \frac{a_0 x^{2k}}{(2.4.6 \cdots 2k)(3.7.11 \cdots (2k-1))}$$

$$= a_0 \left[1 + \sum_{k=1}^{\infty} \frac{x^{2k}}{(2.4.6 \cdots 2k)(3.7.11 \cdots (2k-1))} \right]$$

$$y_g = c_1 y_1 + c_2 y_2$$

Theorem Let $x = x_0$ be a regular singular point of

$$P(x) y'' + Q(x) y' + R(x) y = 0 \quad (1)$$

where P, Q, R are analytic for $|x - x_0| < \rho$

Let, $\alpha_0 = \lim_{x \to x_0} (x - x_0) \frac{Q(x)}{P(x)}$, $\beta_0 = \lim_{x \to x_0} (x - x_0)^2 \frac{R(x)}{P(x)}$

Let, $f(r) = r(r-1) + \alpha_0 r + \beta_0 = 0$ be the indicial eqn. with roots $r_1 \geq r_2$. If they are real. Then, the equation (1) has two linearly independent solutions. One solution is of the form

$$y_1(x) = \sum_{n=1}^{\infty} a_n(r_1) |x - x_0|^{n+r_1} \quad (a_0 \neq 0)$$

a_n is computed using r_1

and the second solution is:

(i) If $r_1 - r_2 \notin \mathbb{Z}$ (not zero or not a integer),
then $y_2(x) = \sum_{n=1}^{\infty} a_n(r_2) |x - x_0|^{n+r_2}$

(ii) If $r_1 - r_2 = N$, a positive integer, then.

$$y_2(x) = A y_1(x) \cdot \ln|x - x_0| + \sum_{n=0}^{\infty} c_n(r_2) (x - x_0)^{n+r_2}$$

(iii) If $r_1 = r_2$, then

$$y_2(x) = y_1(x) \ln|x - x_0| + \sum_{n=0}^{\infty} b_n(r_1) (x - x_0)^{n+r_1}$$

The coefficients $a_n(r_1)$, $b_n(r_1)$, $c_n(r_2)$ and the constant A can be obtained by substituting the form of the series solution for y in D.E (1). Furthermore, all the power series have radius of convergence at least equal to $\rho_1 < \rho$ where $P(z)=0$ does not have any (real or complex) zeros except $z=x_0$ in the disc $|z-x_0|<\rho_1$

Ex 7 $2x^2y'' + xy' + (x^2-3)y = 0$ $\quad x=0$ R.S.P since

$$\alpha_0 = \lim_{x\to 0} x \cdot \frac{x}{2x^2} = \frac{1}{2}, \quad \beta_0 = \lim_{x\to 0} x^2 \frac{x^2-3}{2x^2} = -\frac{3}{2}$$

Indicial eqn. $f(r) = \underbrace{r(r-1) + \frac{1}{2}r - \frac{3}{2}}_{\substack{r^2 - \frac{r}{2} - \frac{3}{2} = 0 \\ 2r^2 - r - 3 = 0}} = 0 \Rightarrow r_1 = \frac{3}{2} > r_2 = -1$

$\Rightarrow (r+1)(2r-3)=0$ and $r_1 - r_2 = \frac{5}{2} \neq N$

So solutions are in the form $y_1 = \sum_{n=0}^{\infty} a_n(r_1) x^{n+r_1}$ and

$$y_2 = \sum_{n=0}^{\infty} a_n(r_2) x^{n+r_2}$$

For y_1: $y = \sum_{n=0}^{\infty} a_n x^{n+r}$ $\quad y' = \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1}$

$$y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2}$$

$$2x^2 \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2} + x \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1}$$

$$+ (x^2-3) \sum_{n=0}^{\infty} a_n x^{n+r} = 0$$

$$2 \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r} + \sum_{n=0}^{\infty} (n+r) a_n x^{n+r}$$

$$+ \underbrace{\sum_{n=0}^{\infty} a_n x^{n+r+2}}_{\sum_{n=2}^{\infty} a_{n-2} x^{n+r}} - 3 \sum_{n=0}^{\infty} a_n x^{n+r} = 0$$

$$2r(r-1)a_0x^r + 2(r+1)ra_1x^{r+1} + ra_0x^r + (r+1)a_1x^{r+1} - 3a_0x^r -$$

$$3a_1x^{r+1} + \sum_{n=2}^{\infty}\underbrace{\left[2(n+r)(n+r-1)a_n + (n+r)a_n + a_{n-2} - 3a_n\right]}_{=0}x^{n+r} = 0$$

① $\{2r(r-1)+r-3\}\underset{\substack{\neq \\ 0}}{a_0}x^r = 0 \quad \Rightarrow \quad \underbrace{2r(r-1)+r-3=0}_{r_1=3/2 \quad r_2=-1}$ "indicial eqn"

② $[2(1+r)r + (1+r) - 3]a_1 = 0$

③ $[(n+r)(2n+2r-1)-3]a_n = -a_{n-2}$.

For y_1 $\quad r_1 = 3/2$ $\quad$ ② $\Rightarrow a_1 = 0$ $\quad (4(1+r)-3)a_1 = 7a_1 = 0$
$(4(\frac{5}{2})-3)a_1 = 0$.

③ $\Rightarrow a_n = -\dfrac{a_{n-2}}{n(2n+5)}$ $\quad n = 2, 3, \ldots$

Let $a_0 = 1$

$a_{2k+1} = 0 \quad k = 0, 1, 2, \ldots$

$a_{2k} = \dfrac{(-1)^k a_0}{2\cdot4\ldots2k\cdot9\cdot13\ldots(4k+5)} \quad k = 1, 2, \ldots$

$$y_1(x) = \sum_{k=0}^{\infty} a_n(r_1)x^{n+3/2} = \underbrace{a_0(\tfrac{3}{2})}_{1}\cdot x^{3/2} + \sum_{k=1}^{\infty}\frac{(-1)^k a_0}{2\cdot4\ldots2k\ 9\cdot13\ldots(4k+5)}x^{2k+3/2}$$

$$y_1(x) = \underset{\substack{\| \\ 1}}{a_0}|x|^{3/2}\left(1 + \sum_{k=1}^{\infty}\frac{(-1)^k}{2\cdot4\ldots(2k)\ 9\cdot13\ldots(4k+5)}x^{2k}\right)$$

For other solution ② $r=-1$ to get $-3a_1 = 0 \Rightarrow a_1 = 0$

③ $\Rightarrow a_n = -\dfrac{a_{n-2}}{n(2n-5)}$ $\quad n = 2, 3, \ldots$

$k = 1, 2, \ldots$

$a_{2k+1} = 0$ and $a_{2k} = \dfrac{(-1)^k a_0}{2\cdot4\ldots(2k)(-1)3\ldots(4k-5)}$,

$$y_2(x) = |x|^{-1}\left[1 + \sum_{k=1}^{\infty}\frac{(-1)^k}{2\cdot4\ldots(2k)\cdot(-1)\cdot3\ldots(4k-5)}x^{2k}\right]$$

with $a_0 = 1$ $\qquad y_{general} = c_1y_1 + c_2y_2$.

Ex $x^2y'' + xy' + x^2y = 0.$

$x=0$ is a RSP since $\alpha_0 = \lim_{x\to 0} x \cdot \frac{x}{x^2} = 1$ and $\beta_0 = \lim_{x\to 0} x^2 \frac{x^2}{x^2}$

$f(r) = r(r-1) + 1.r + 0 = 0 \Rightarrow r^2 = 0 \Rightarrow r_{1,2} = 0 \quad r_1 = r_2 = 0$

$y = \sum_{n=0}^{\infty} a_n x^{n+r}, \quad y' = \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1}, \quad y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2}$

$$\sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r} + \sum_{n=0}^{\infty} (n+r) a_n x^{n+r} + \underbrace{\sum_{n=0}^{\infty} a_n x^{n+r+2}}_{\sum_{n=2}^{\infty} a_{n-2} x^{n+r}} = 0$$

$$r(r-1) a_0 x^r + (r+1) r a_1 x^{r+1} + r a_0 x^r + (r+1) a_1 x^{r+1}$$
$$+ \sum_{n=2}^{\infty} \left([(n+r)(n+r-1) + n+r] a_n + a_{n-2} \right) x^{n+r} = 0$$

① $r(r-1) + r = 0$ "indicial eqn" $a_0 \neq 0$

② $[(r+1) r + (r+1)] a_1 = 0.$

③ $(n+r)(n+r) a_n = -a_{n-2} \quad n \geq 2$

Let $r_1 = r = 0$

② $\Rightarrow a_1 = 0$

③ $n^2 a_n = -a_{n-2} \quad n \geq 2$

$a_n = -\frac{1}{n^2} a_{n-2}$

$\vdots$

$a_{2k} = \frac{(-1)^k a_0}{2^2 4^2 \cdots (2k)^2} \quad k \geq 1 \quad a_{2k+1} = 0$

$$y_1(x) = \sum_{n=0}^{\infty} a_n x^{n+r} = \sum_{n=0}^{\infty} a_n x^n = a_0 + \sum_{k=1}^{\infty} \frac{(-1)^k x^{2k}}{(2^k k!)^2} a_0$$

$$y_1(x) = a_0 \left[1 + \sum_{k=1}^{\infty} \frac{(-1)^k x^{2k}}{(2^k k!)^2} \right]$$

To find $y_2(x)$, we could try to substitute

$$y_2 = y_1(x) \ln x + \sum_{n=0}^{\infty} b_n x^n$$

but instead we'll try a different way as in Euler eqn.

Ignore the index $r=0$ and solve the reccurence relation in terms of r, assuming $r\neq 0$. This gives:

$$\{r(r+1)+(r+1)\}a_1 = (r+1)^2 a_1 = 0 \Rightarrow a_1 = 0$$

$$\frac{d}{dr}(x^{n+r}) = x^{n+r}\ln x$$

$$(n+r)^2 a_n = -a_{n-2} \quad n\geq 2.$$

$$a_n = -\frac{a_{n-2}}{(n+r)^2}$$

$$a_1 = a_3 = \cdots = a_{2k+1} = 0 \qquad a_{2k} = \frac{(-1)^k a_0}{(2+r)^2(4+r)^2\cdots(2k+r)^2}$$

$$y_2(x) = \frac{\partial}{\partial r}\Big[\sum_{n=0}^{\infty} a_n(r)x^{n+r}\Big]\Big|_{r=0}$$

$$= \Big[\sum_{n=0}^{\infty} a_n'(r)x^{n+r} + \sum_{n=0}^{\infty} a_n(r)x^{n+r}\ln x\Big]\Big|_{r=0}.$$

derivative $= \frac{2}{2+r}$

$$a_n(r) = \frac{(-1)^k a_0}{(2+r)^2(4+r)^2\cdots(2k+r)^2} \quad \ln(2+r)^2 + \ln(4+r)^2 + \cdots + \ln(2k+r)^2$$

$$\ln a_n(r) = \ln(-1)^k a_0 - \ln\big((2+r)^2\cdots(2k+r)^2\big) \quad \text{diff.}$$

w.r.t. r $\quad \frac{d}{dr}(\ln(a_n(r))) = \frac{1}{a_n(r)}\frac{d\,a_n(r)}{dr} \Rightarrow \frac{d\,a_n(r)}{dr} = a_n(r)\frac{d}{dr}\ln a_n$

$$y_2(x) = \frac{\partial}{\partial r}\Big[x^r\Big(1+\sum_{k=1}^{\infty}\frac{(-1)^k x^{2k}}{(2+r)^2(4+r)^2\cdots(2k+r)^2}\Big)\Big]\Big|_{r=0}$$

$y_1(x)$ $\quad a_n(r)$ $\quad$ For simplic. $a_0 = 1$

$$= x^r\ln x\Big[1+\sum_{k=1}^{\infty}\frac{(-1)^k x^{2k}}{(2+r)^2(4+r)^2\cdots(2k+r)^2}\Big]$$

$$+\, x^r\Big[\sum_{k=1}^{\infty}(-1)^k x^{2k}\underbrace{\frac{\partial}{\partial r}\Big(\frac{1}{(2+r)^2(4+r)^2\cdots(2k+r)^2}\Big)}_{\frac{-2}{(2+r)^2\cdots(2k+r)^2}\left[\frac{1}{2+r}+\frac{1}{4+r}+\cdots+\frac{1}{2k}\right]}\Big]$$

[Euler equation same root
$y_2(x) = \frac{\partial}{\partial x}[y_1 \ln x]\big|_{r=r_1}$]

Thus, $y_2\big|_{r=0} = \ln x\Big[1+\sum_{k=1}^{\infty}\frac{(-1)^k x^{2k}}{2^2 4^2\cdots(2k)^2}\Big]$

$$-\sum_{k=1}^{\infty}\frac{(-1)^k}{2^2 4^2\cdots(2k)^2}\Big(\frac{1}{2}+\frac{1}{4}+\cdots+\frac{1}{2k}\Big)x^{2k}$$

Since we have double root $r_1=r_2=0$

$$y_2(x) = y_1\ln x + \sum_{k=1}^{\infty}\frac{(-1)^{k+1}}{2^{2k}(k!)^2}\Big(1+\frac{1}{2}+\cdots+\frac{1}{k}\Big)x^{2k}$$

$\frac{1}{2k}$